Task 2: Access Control

Task 2a

Introduction

This report aims to detail the process of an administrator with sudo rights configuring a share-folder and other subfolders under it. It will also show how users with different permissions can access the system.

The user which will be performing these steps are on with administrator rights (User: fit2093).

New User Groups

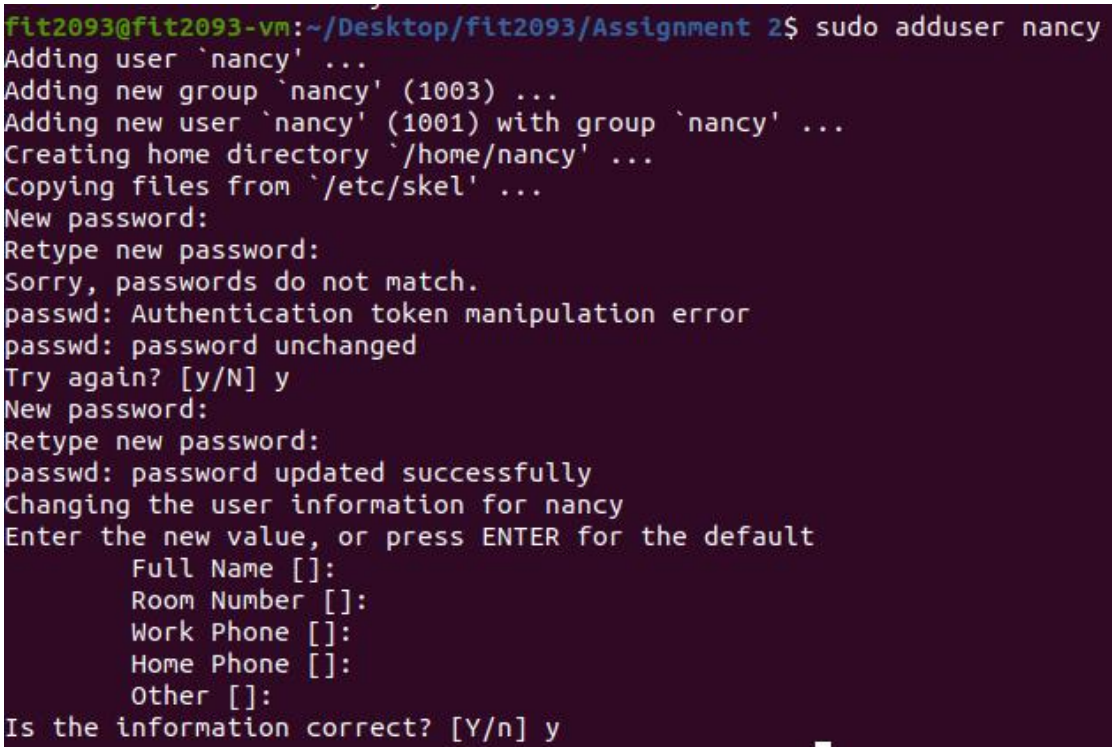
Two new user groups Human Resources (hr) and Public Relations (pr) are created using the “addgroup” command.

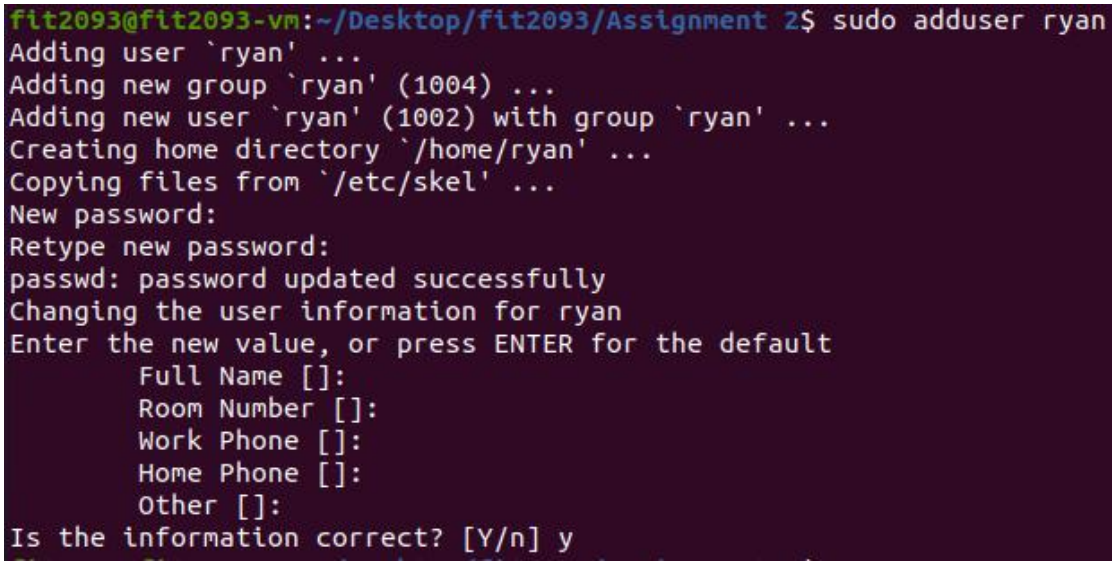
```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo addgroup hr
[sudo] password for fit2093:
Adding group `hr' (GID 1001) ...
Done.
```

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo addgroup pr
Adding group `pr' (GID 1002) ...
Done.
```

New Users

Two new users “nancy” and “ryan” are created using the “adduser” command.





Next, “nancy”, who is in both groups “hr” and “pr”, and “ryan”, who is only in group “pr”, are added to their respective groups using the “adduser” command.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo adduser nancy hr
Adding user `nancy' to group `hr' ...
Adding user nancy to group hr
Done.
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo adduser nancy pr
Adding user `nancy' to group `pr' ...
Adding user nancy to group pr
Done.
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo adduser ryan pr
Adding user `ryan' to group `pr' ...
Adding user ryan to group pr
Done.
```

Contents

Using the command below, it can be observed that the users are correctly added to their respective groups.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$  cat /etc/group
```

Output:

```
hr:x:1001:nancy
pr:x:1002:nancy,ryan
nancy:x:1003:
ryan:x:1004:
```

Nancy has access to both “hr” and “pr” as well as her home directory “nancy” while Ryan has access to only “pr” and his home directory “ryan”.

Task 2b

Folders and Access

As a user with administrator rights (fit2093), folders can be created, and their permissions can be set.

First, login as fit2093 (admin).

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo su fit2093
[sudo] password for fit2093:
```

The share-folder is created with the command below.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo mkdir /home/share-folder
```

The subfolders pr and hr are created with the commands below.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo mkdir /home/share-folder/pr
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo mkdir /home/share-folder/hr
```

The groups pr and hr are then assigned to their corresponding folders with the commands below.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo chgrp pr /home/share-folder/pr
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2$ sudo chgrp hr /home/share-folder/hr
```

The permissions for these groups are set using the commands below. 770 means that the owner and group member can read, write and execute while anyone else has no access to these folders.

```
fit2093@fit2093-vm:/home/share-folder$ sudo chmod 770 /home/share-folder/pr
fit2093@fit2093-vm:/home/share-folder$ sudo chmod 770 /home/share-folder/hr
```

User Access

The outcome of Nancy and Ryan accessing the two folders is shown below. Nancy is a group member of both hr and pr while Ryan is only a group member of pr.

First, Nancy logs in.

```
fit2093@fit2093-vm:/home/share-folder$ sudo su nancy
```

Nancy creates a text file in the hr folder using the command below.

```
nancy@fit2093-vm:/home/share-folder$ echo "nancyhr" > /home/share-folder/hr/nancyhr.txt
```

Next, Ryan logs in.

```
nancy@fit2093-vm:/home/share-folder$ su ryan
```

Ryan creates a text file in the pr folder using the command below.

```
ryan@fit2093-vm:/home/share-folder$ echo "ryanpr" > /home/share-folder/pr/ryanpr.txt
```

When Ryan attempts to open the files, he is able to open the file in pr but is denied access to the file in hr.

```
ryan@fit2093-vm:/home/share-folder$ cat /home/share-folder/pr/ryanpr.txt
ryanpr
```

```
ryan@fit2093-vm:/home/share-folder$ cat /home/share-folder/hr/nancyhr.txt
cat: /home/share-folder/hr/nancyhr.txt: Permission denied
```

When Nancy attempts to open the files, she is able to open both the files in pr and hr.

```
ryan@fit2093-vm:/home/share-folder$ su nancy
```

```
nancy@fit2093-vm:/home/share-folder$ cat /home/share-folder/pr/ryanpr.txt
ryanpr
```

```
nancy@fit2093-vm:/home/share-folder$ cat /home/share-folder/hr/nancyhr.txt
nancyhr
```

Task 2c

Readsecret Permissions

First, the readsecret_x86.tar.gz is copied to the share-folder created earlier.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2/2c$ sudo cp readsecret_x86.tar.gz /home/share-folder/
```

Using the commands below, the directory is changed to the share-folder and its contents are displayed.

```
fit2093@fit2093-vm:~/Desktop/fit2093/Assignment 2/2c$ cd /home/share-folder
```

```
fit2093@fit2093-vm:/home/share-folder$ ls -la
total 20
drwxr-xr-x 4 root root 4096 May 21 00:41 .
drwxr-xr-x 6 root root 4096 May 20 23:48 ..
drwxrwx--- 2 root hr   4096 May 21 00:15 hr
drwxrwx--- 2 root pr   4096 May 21 00:18 pr
-rw-r--r-- 1 root root 3035 May 21 00:41 readsecret_x86.tar.gz
```

The file is executed which contains the following files:

```
fit2093@fit2093-vm:/home/share-folder$ sudo tar -zxvf readsecret_x86.tar.gz
Asg2/
Asg2/directory/
Asg2/directory/secret.txt
Asg2/readsecret
```

As seen below, “directory” can only be accessed by the owner which includes any file under it such as “secret.txt”. “readsecret” can be accessed by the owner and those of group “fit2093” but anyone else can only read and execute it.

```
fit2093@fit2093-vm:/home/share-folder$ cd /home/share-folder/Asg2
fit2093@fit2093-vm:/home/share-folder/Asg2$ ls -la
total 32
drwxrwxr-x 3 fit2093 fit2093  4096 Apr 23 20:05 .
drwxr-xr-x 5 root    root     4096 May 21 00:44 ..
drwx------ 2 fit2093 fit2093  4096 Apr 23 19:36 directory
-rwxrwxr-x 1 fit2093 fit2093 16880 Apr 23 19:58 readsecret
```

SUID permission for “readsecret” using the command below.

```
fit2093@fit2093-vm:/home/share-folder/Asg2$ chmod u+s /home/share-folder/Asg2/readsecret
```

```
fit2093@fit2093-vm:/home/share-folder/Asg2$ ls -la
total 32
drwxrwxr-x 3 fit2093 fit2093  4096 Apr 23 20:05 .
drwxr-xr-x 5 root    root     4096 May 21 00:44 ..
drwx------ 2 fit2093 fit2093  4096 Apr 23 19:36 directory
-rwsrwxr-x 1 fit2093 fit2093 16880 Apr 23 19:58 readsecret
```

Readsecret Outcomes

This is what happens when users with different permissions directly access “secret.txt” and when they run “readsecret”

FIT2093 (fit2093):

```
fit2093@fit2093-vm:/home/share-folder/Asg2$ cat /home/share-folder/Asg2/directory/secret.txt
This is a file for Asg 2 that you should not read it without running readsecret
```

```
fit2093@fit2093-vm:/home/share-folder/Asg2$ ./readsecret directory/secret.txt
Congratulations! You get it!
This is a file for Asg 2 that you should not read it without running readsecret
```

Nancy (pr & hr):

```
fit2093@fit2093-vm:/home/share-folder/Asg2$ sudo su nancy
```

```
nancy@fit2093-vm:/home/share-folder/Asg2$ cat /home/share-folder/Asg2/directory/secret.txt
cat: /home/share-folder/Asg2/directory/secret.txt: Permission denied
```

```
nancy@fit2093-vm:/home/share-folder/Asg2$ ./readsecret directory/secret.txt
Congratulations! You get it!
This is a file for Asg 2 that you should not read it without running readsecret
```

Ryan (pr):

```
nancy@fit2093-vm:/home/share-folder/Asg2$ su ryan
```

```
ryan@fit2093-vm:/home/share-folder/Asg2$ cat /home/share-folder/Asg2/directory/secret.txt
cat: /home/share-folder/Asg2/directory/secret.txt: Permission denied
```

```
ryan@fit2093-vm:/home/share-folder/Asg2$ ./readsecret directory/secret.txt
Congratulations! You get it!
This is a file for Asg 2 that you should not read it without running readsecret
```

As observed above, while everyone can run “readsecret” to gain access to “secret.txt”, only those in the group “fit2093” can directly access “secret.txt”

Conclusion

In conclusion, by setting permissions and giving certain users the ability to interact with only specific parts of a system, we can maintain the security of the system by grating a user’s access to only what they are allowed to.